Passive buzzer

DO PICO SOUND PWM

What does it do?

A passive buzzer needs fast pulses (PWM) to make sound. Changing their frequency changes its pitch: low or high notes.

Recognise it: check for **passive**, **external drive** or **no oscillator** on the packet. Both types can look identical; + only marks polarity.

Use a small **3.3 V GPIO-compatible piezo** sounder.

Uses



Washing-machine chime

PWM plays tunes; active buzzers have only one pitch.

DETECT Read cycle status → **DECIDE** Has the wash finished? → **DO** Play a short tune



Robot voice box

Change pitch for happy chirps or grumpy growls.

DETECT Read mood button → **DECIDE** Is happy pressed? → **DO** Chirp high or low



+ leg
Marked +, to GPIO

− leg
Other leg, to GND

Vent
Sound comes out

Raspberry Pi Foundation

Code

rpf.io/soundfx

Raspberry Pi Pico

```
from picozero import PWMBuzzer

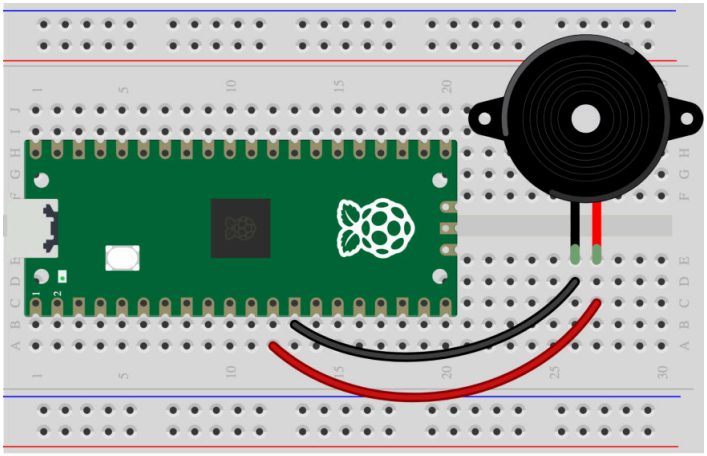
# GP9, 2000 Hz, 50% duty
buzzer = PWMBuzzer(9, 2000, 32768)

buzzer.on()
```

Try `buzzer.freq = 1000`; silence with `buzzer.off()`. Higher frequency means higher pitch. Beep speed is separate; both types can make rhythms.

Wiring

PICO



GP9 (pin 12) → **+** buzzer **–** → **GND** (pin 13)